

1. Indicate whether the following statements are true or false. Justify your answers.

- a) Real-time CPS applications usually require a response within a very short time.
- b) A guest OS abstracts H/W and provides an interface between H/W and hypervisor.
- c) Managing the interactions between a guest OS and its applications is one of the key functions of a hypervisor.
- d) Under a Type-1 para-virtualized hypervisor, the hypervisor directly interacts with the H/W and provides a virtual call API to interact with the guest OS.

1a) Real-time CPS applications usually require a response within a very short time.

→ False.

Justification: Real-time does not mean fast or in a very short time. It means within bounded time in the worst-case. So, the response times for such applications have to be guaranteed within a pre-defined deadline in the worst-case.

1b) A guest OS abstracts H/W and provides an interface between H/W and hypervisor.

→ False.

Justification: A **host OS** abstracts H/W and provides an interface between H/W and hypervisor under type-2 virtualization. The guest OS interacts with the hypervisor and provides services to the applications running on it.

1c) Managing the interactions between a guest OS and its applications is one of the key functions of a hypervisor.

→ False.

Justification: Managing the interactions between the guest OS and the H/W is the key function of a hypervisor. It does so through its virtual machine management framework. The interactions between the guest OS and its applications are completely independent of the hypervisor.

1d) Under a Type-1 para-virtualized hypervisor, the hypervisor directly interacts with the H/W and provides a virtual call API to interact with the guest OS.

→ True.

Justification: Under Type-1 or bare metal virtualization, the hypervisor interacts directly with hardware. Since the hypervisor is also para-virtualized, not all of the H/W instructions are virtualized. Therefore, the guest OS needs to be modified to interact with the hypervisor and this interaction will use the virtual call API.

2. You are tasked with setting up a virtualized environment to run a variety of workloads, including data processing (heavyweight) applications and web servers (lightweight), on a number of physical servers.

The goal is to maximize performance and resource utilization while ensuring flexibility across different workloads.

Emulation:

- **Explanation:** Emulation allows running applications or operating systems that were designed for different hardware or environments. This can be crucial when dealing with legacy applications.
- **Application:** In this case, if an application or operating system (e.g., a legacy data processing system) requires different hardware, emulation can allow it to run on modern servers.
- **Challenge:** Emulation can introduce performance overhead due to the need for hardware instruction translation, slowing down time-sensitive workloads like real-time data processing.

Multiplexing:

- **Explanation:** Multiplexing involves sharing a physical resource (such as a CPU core or memory) across multiple virtual machines or applications by allocating them time slices.
- **Application:** Multiplexing would be particularly useful for handling lightweight workloads, such as web servers, that don't need continuous CPU or memory access. The server can allocate time slices to each workload, increasing resource utilization.
- **Challenge:** Proper scheduling is critical. Overcommitting resources or poor allocation strategies can lead to performance bottlenecks, especially for workloads like heavy data processing that require more cores and memory.

Aggregation:

- **Explanation:** Aggregation pools multiple physical resources (e.g., multiple CPU cores or memory modules) into one large virtual resource.
- **Application:** Aggregation is useful for scaling up resource-intensive tasks, like heavy data processing workloads that require more computational power than a single server can provide.
- **Challenge:** Aggregation might reduce resource availability for smaller workloads (e.g., web servers) and increase the complexity of managing those resources. Ensuring that resources are efficiently distributed and isolated can be a challenge.

Recommendation: The tutor should guide students to understand the trade-offs based on the type of workloads:

- **Multiplexing:** Useful for lightweight workloads (e.g., web servers) that don't require continuous access to resources.
- **Aggregation:** Beneficial for large-scale applications (e.g., database servers or data processing) that need pooled resources to handle bigger tasks.
- **Emulation:** Useful for running legacy or incompatible applications, but may introduce significant performance overhead.
- Tutors should encourage students to prioritize based on workload requirements. For instance, web servers might benefit from multiplexing, while data processing might require aggregation for scalability.

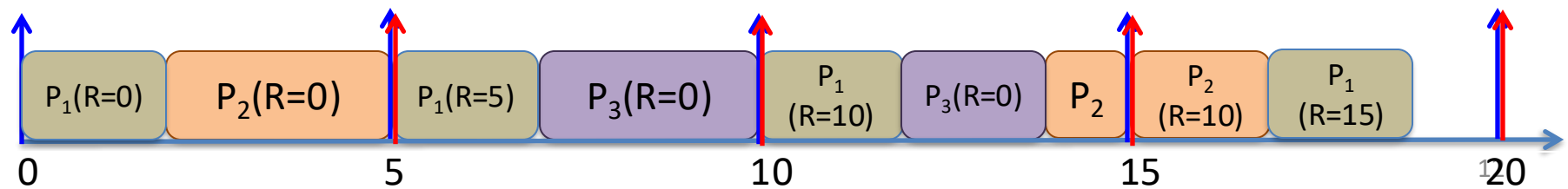
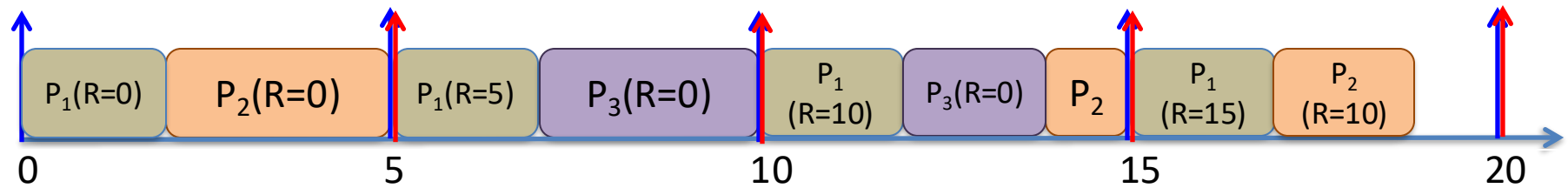
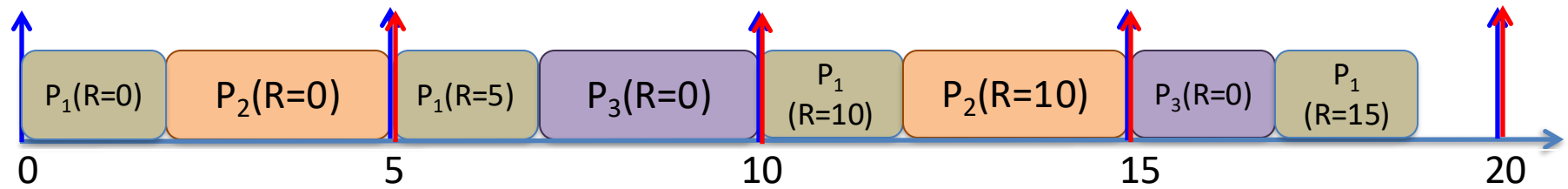
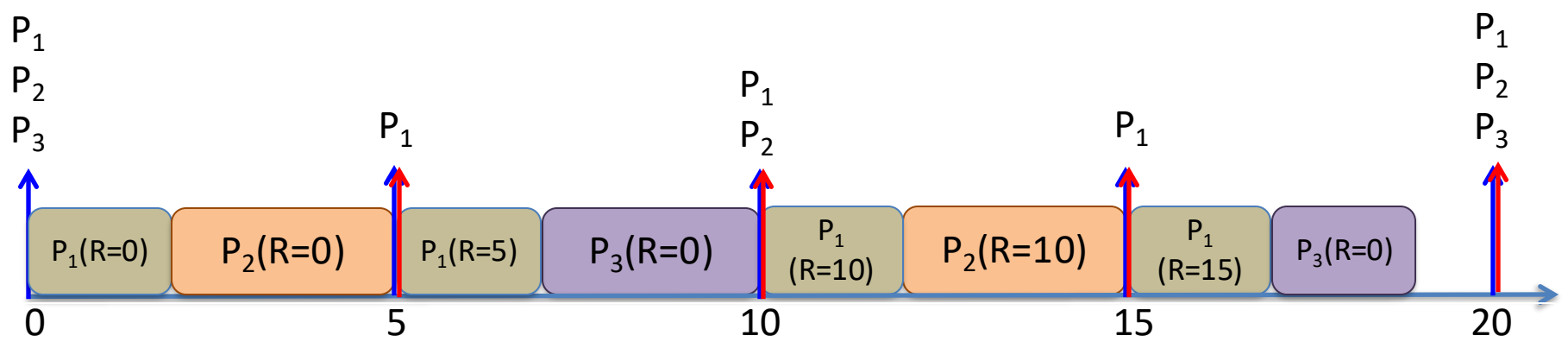
3. Consider the following set of periodic real-time processes:

$P1\langle 5, 2, 5 \rangle$

$P2\langle 10, 3, 10 \rangle$

$P3\langle 20, 5, 20 \rangle$

Construct all possible schedules for this process set under the earliest deadline first (EDF) scheduling strategy. Is this process set schedulable under EDF?



- There are four possible schedules under EDF
 - Tie break between $P_1(R=15)$ and $P_3(R=0)$
 - Tie break between $P_2(R=10)$ and $P_3(R=0)$
 - Tie break between $P_1(R=15)$ and $P_2(R=10)$
- Deadline is met under all the above schedules; the process set is therefore schedulable under EDF